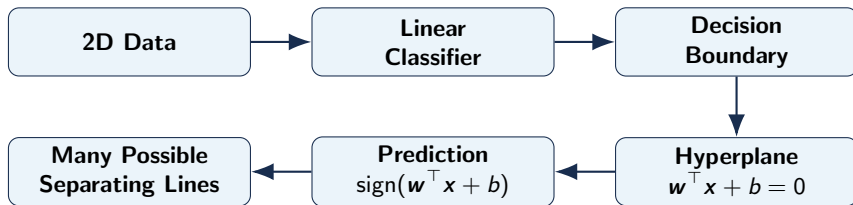


Support Vector Machines

Classification Using a Linear Boundary

Machine Learning

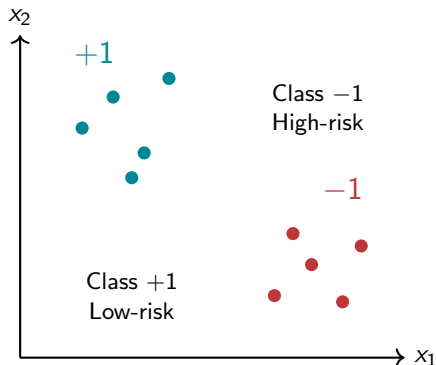
Where Are We Going?



Big Idea

Before learning the “support vector” part of SVM, we first need to understand how a linear classifier separates two classes using a straight line.

8. Two-Dimensional Classification Example



Example

Suppose every observation has two features:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

For example:

x_1 = income score

x_2 = repayment score.

The class label is

$$y \in \{-1, +1\}.$$

9. Linear Classifier



Linear Classifier

A linear classifier first calculates a score:

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$$

For two features,

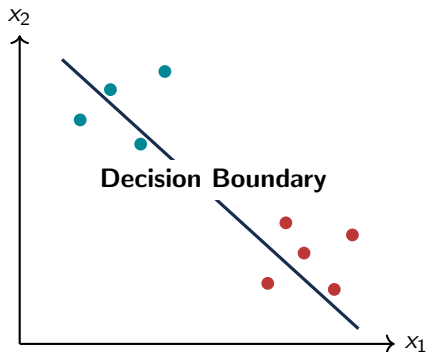
$$f(\mathbf{x}) = w_1 x_1 + w_2 x_2 + b.$$

Then:

$$f(\mathbf{x}) > 0 \Rightarrow \text{Class } +1$$

$$f(\mathbf{x}) < 0 \Rightarrow \text{Class } -1.$$

10. Decision Boundary



What is the decision boundary?

The decision boundary is where the classifier is exactly at zero:

$$f(\mathbf{x}) = 0.$$

Therefore,

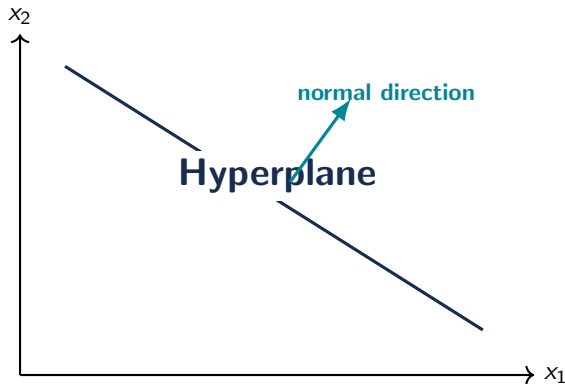
$$\mathbf{w}^\top \mathbf{x} + b = 0$$

This equation divides the feature space into two regions.

$$f(\mathbf{x}) > 0 \Rightarrow +1$$

$$f(\mathbf{x}) < 0 \Rightarrow -1$$

11. Hyperplane



Important terminology

- In 2 dimensions: a hyperplane is a **line**.
- In 3 dimensions: a hyperplane is a **plane**.
- In p dimensions: a hyperplane has dimension $p - 1$.

12. Equation of a Hyperplane

$$\mathbf{w}^\top \mathbf{x} + b = 0$$

Numerical Example

Let

$$\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad b = -4.$$

Then

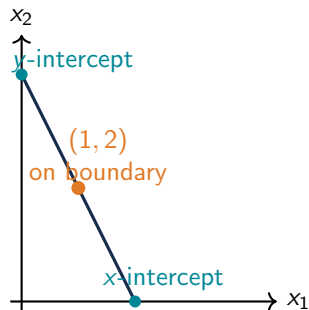
$$\mathbf{w}^\top \mathbf{x} + b = 2x_1 + x_2 - 4.$$

Therefore the boundary is

$$2x_1 + x_2 - 4 = 0$$

or

$$x_2 = 4 - 2x_1$$



12. Numerical Example: Checking Points

Given

The decision boundary is

$$2x_1 + x_2 - 4 = 0.$$

We will test different observations.

Point A:

$$\mathbf{x}_A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

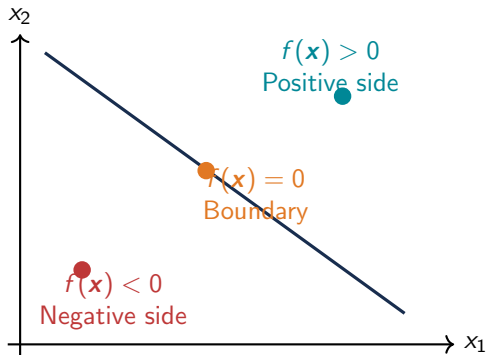
Substitute:

$$2(1) + 2 - 4 = 0.$$

Therefore,

$(1, 2)$ lies on the boundary.

Three Possible Situations for a Point



$$f(\mathbf{x}) < 0 \Rightarrow -1$$

$$f(\mathbf{x}) = 0 \Rightarrow \text{boundary}$$

$$f(\mathbf{x}) > 0 \Rightarrow +1$$

13. Role of w and b

Role of w : Orientation

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

controls the **orientation** of the decision boundary.

$$w_1x_1 + w_2x_2 + b = 0$$

$$\mathbf{w} \perp \text{boundary}$$

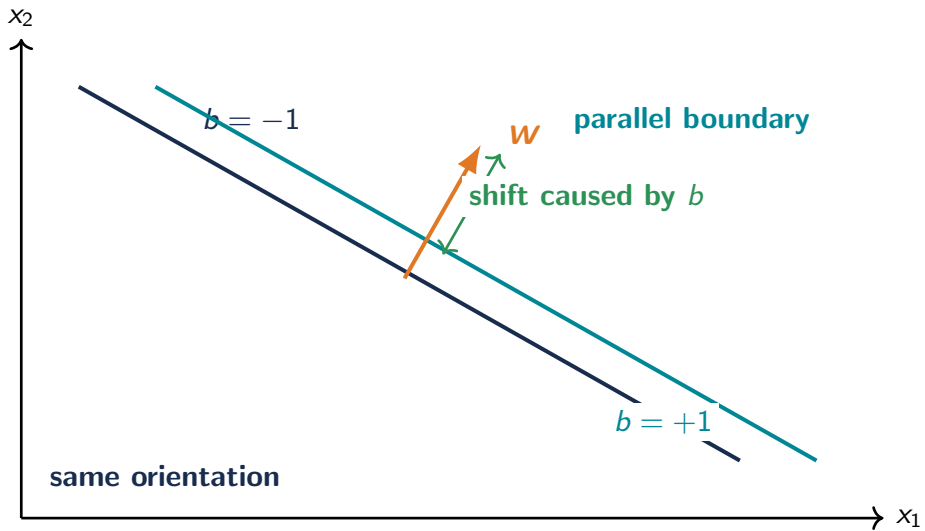
Role of b : Position

The intercept b controls the **position** of the boundary.

Changing b shifts the line while keeping its orientation unchanged.

$$w_1x_1 + w_2x_2 + b = 0$$

$$\mathbf{w} \text{ unchanged} \Rightarrow \text{same orientation}$$



$w \longrightarrow$ orientation

$b \longrightarrow$ position

13. Numerical Example: Changing w and b

Starting boundary

$$2x_1 + x_2 - 4 = 0$$

or

$$x_2 = 4 - 2x_1.$$

Change b only

$$2x_1 + x_2 - 2 = 0$$

Therefore,

$$x_2 = 2 - 2x_1.$$

Same orientation

but

different position.

Change w

Take

$$w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Then

$$x_1 + 2x_2 - 4 = 0$$

and

$$x_2 = 2 - \frac{1}{2}x_1.$$

Different orientation.

14. Classification Rule

$$\hat{y} = \text{sign}(\mathbf{w}^\top \mathbf{x} + b)$$

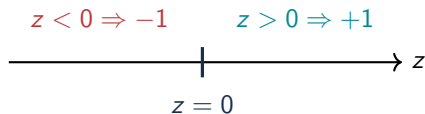
What does sign mean?

$$\text{sign}(z) = \begin{cases} +1, & z > 0, \\ -1, & z < 0. \end{cases}$$

At

$$z = 0,$$

the point lies on the decision boundary.



14. Worked Numerical Classification Example

Given Classifier

$$\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad b = -4$$

Therefore,

$$\hat{y} = \text{sign}(2x_1 + x_2 - 4)$$

Point A

$$\mathbf{x}_A = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Calculate the score:

$$f(\mathbf{x}_A) = 2(3) + 2 - 4 = 6 + 2 - 4 = 4.$$

Since $4 > 0$, the point lies on the positive side.

$$\hat{y}_A = +1$$

Point B

$$\mathbf{x}_B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

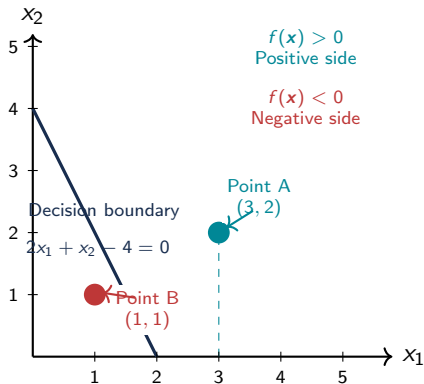
Calculate the score:

$$f(\mathbf{x}_B) = 2(1) + 1 - 4 = 2 + 1 - 4 = -1.$$

Since $-1 < 0$, the point lies on the negative side.

$$\hat{y}_B = -1$$

Visual Interpretation of the Classification



Point A

$$\mathbf{x}_A = (3, 2)$$

$$f(\mathbf{x}_A) = 4, \quad f(\mathbf{x}_A) > 0$$

$$\hat{y}_A = +1$$

Positive side

Point B

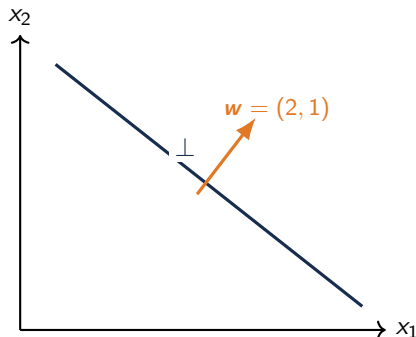
$$\mathbf{x}_B = (1, 1), \quad f(\mathbf{x}_B) = -1$$

$$f(\mathbf{x}_B) < 0$$

$$\hat{y}_B = -1$$

Negative side

15. Geometric Interpretation of w



Why is w perpendicular?

Take two points x_1 and x_2 on the boundary.
Then

$$w^T x_1 + b = 0$$

and

$$w^T x_2 + b = 0.$$

Subtract:

$$w^T (x_1 - x_2) = 0.$$

Thus the direction along the boundary is orthogonal to w .

15. Numerical Proof That $\mathbf{w}$ Is Normal

Boundary

Consider $2x_1 + x_2 - 4 = 0$. Therefore, $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Choose two points on the boundary:

$$A = (0, 4), \quad B = (2, 0).$$

The direction along the line is

$$B - A = (2, -4).$$

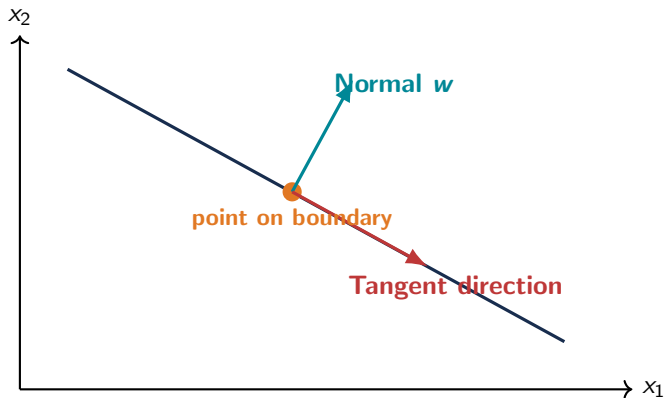
Now calculate the dot product:

$$\mathbf{w}^\top (B - A) = (2, 1) \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2(2) + 1(-4) = 4 - 4 = \boxed{0}.$$

Hence,

$$\mathbf{w} \perp \text{boundary direction.}$$

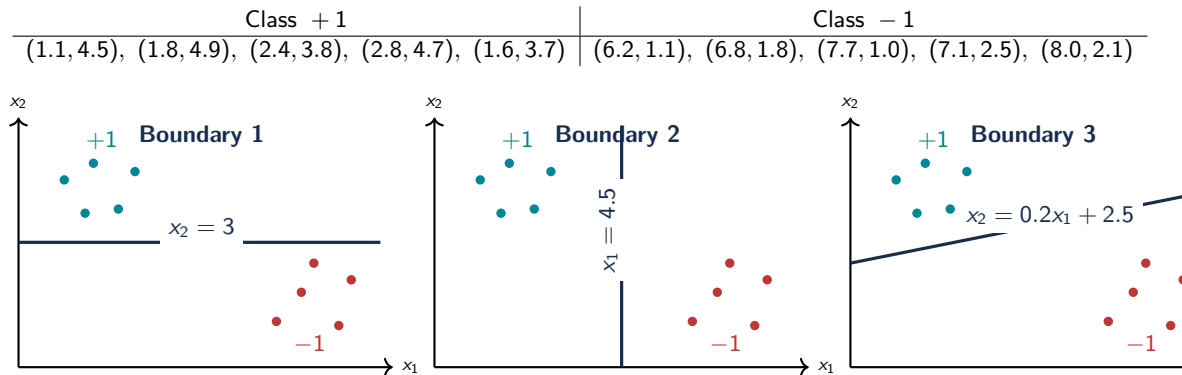
Geometric Picture: Tangent vs Normal



$$\text{Normal vector} \cdot \text{tangent vector} = 0$$

Why Can Multiple Decision Boundaries Exist?

Consider the same training data:



All three boundaries correctly separate the two classes.

Different separating boundaries can have zero training error.

Therefore, SVM needs another principle to choose the **best** boundary.

16. Numerical Comparison of Two Separating Lines

Suppose we have four observations:

Class + 1	Class - 1
(2, 4)	(6, 1)
(3, 3.2)	(7, 1.5)

Candidate A

$$x_2 = 5 - 0.65x_1.$$

The line separates the two classes.

Training errors = 0.

Candidate B

$$x_2 = 4 - 0.45x_1.$$

This line also separates the two classes.

Training errors = 0.

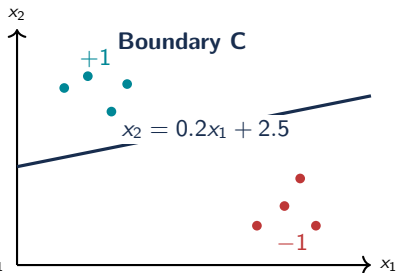
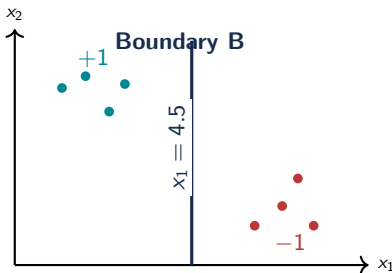
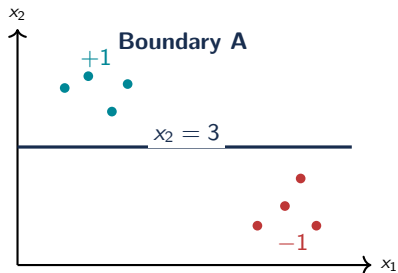
Which line should we choose?

Coming next

SVM introduces a geometric criterion: Choose the separating boundary with the largest margin.

Many Boundaries $\neq$ One Answer

The **same dataset** can be separated correctly by different decision boundaries.



The SVM Question

All three boundaries classify the displayed training observations correctly:

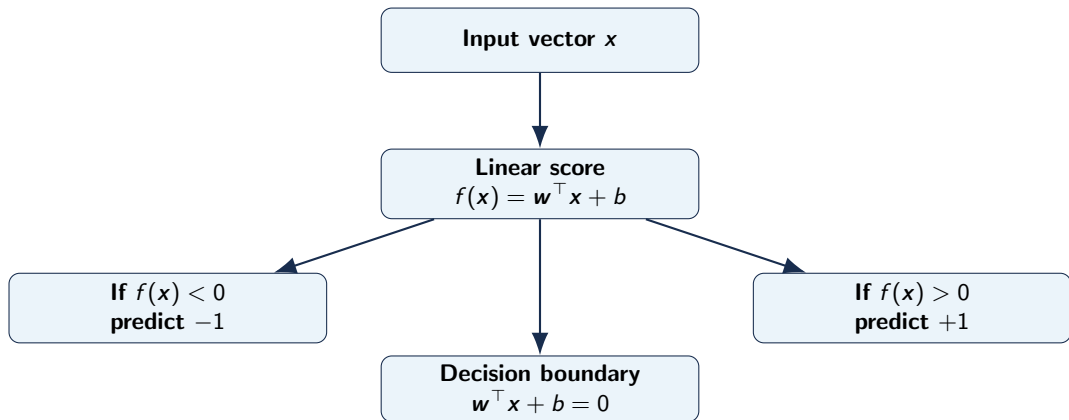
Training accuracy = 100%

But the boundaries are different. Which boundary should we choose?



SVM chooses using the maximum-margin principle.

Putting Everything Together



The complete linear-classification idea

$$x \longrightarrow w^T x + b \longrightarrow \text{sign} \longrightarrow \hat{y}$$

Quick Check for Students

Try these before moving to the margin concept.

① For

$$\mathbf{w} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \quad b = 6,$$

write the decision boundary.

② Classify

$$\mathbf{x} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

using

$$\hat{y} = \text{sign}(3x_1 - 2x_2 + 6).$$

③ What happens to the boundary if b changes?

④ Why is $\mathbf{w}$ perpendicular to the boundary?

⑤ Why can several separating lines have zero training error?

Quick Check: Answers

1. Decision boundary

$$3x_1 - 2x_2 + 6 = 0.$$

Therefore,

$$x_2 = \frac{3}{2}x_1 + 3$$

2. Classification

For $(0, 2)$,

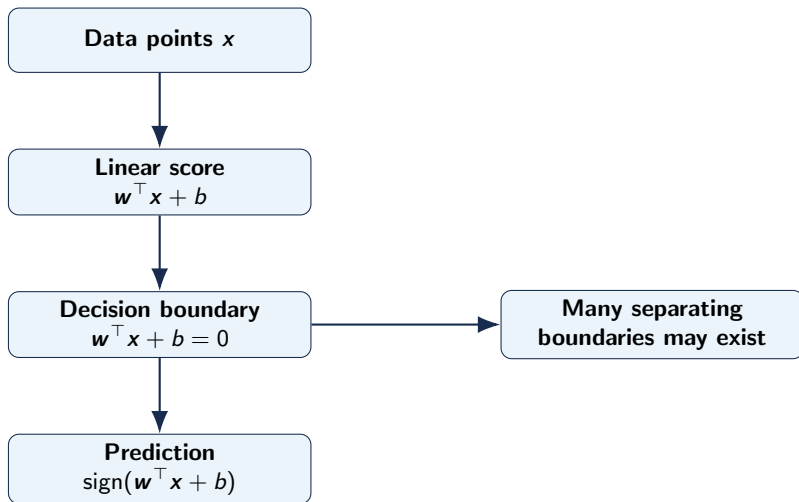
$$3(0) - 2(2) + 6 = 2.$$

Since

$$2 > 0,$$

$$\hat{y} = +1.$$

Summary: Classification Using a Linear Boundary



The Question That Leads Us to SVM

Suppose many different lines classify the training observations correctly.

Which one is best?



Maximum Margin